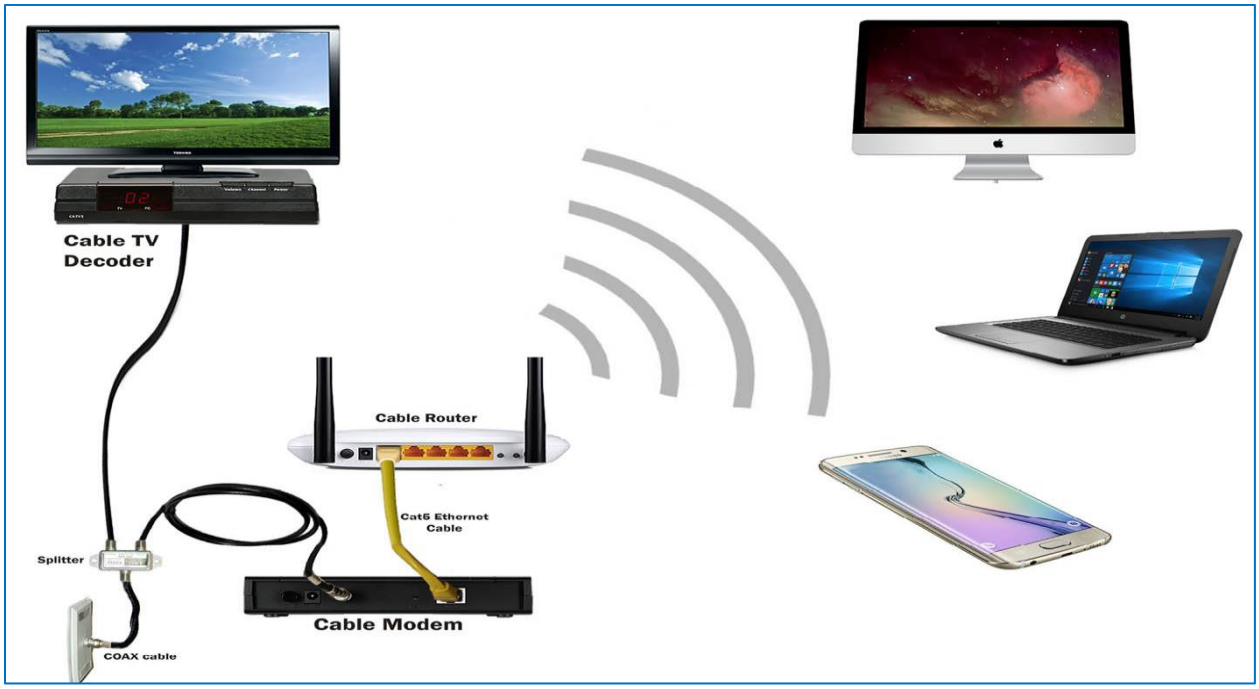
Assignment

Module 3.1 Introduction to Networking hardware Theory

1. **Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.**

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2. **List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').**

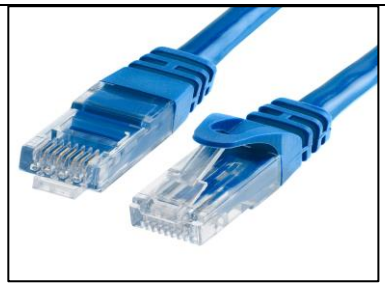
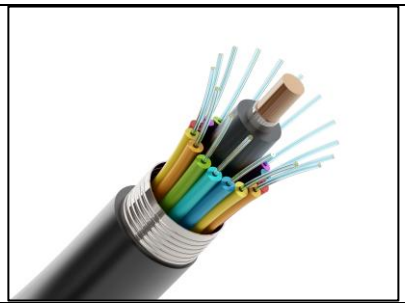
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Networking Device	Main Function	Real-Life Example
Modem	Connects a home or office network to the internet service provider (ISP).	At home, connect to the internet through a broadband connection.
Router	Directs data between different networks and shares an internet connection among multiple devices.	At home, to share the internet with phones, laptops, and smart TVs.
Switch	Connects multiple devices within a local area network (LAN) and forwards data to the correct device.	In a school computer lab, connect multiple computers to the same network.
Access Point	Provides Wi-Fi connectivity so wireless devices can connect to a wired network.	In an office, to allow employees to connect their laptops and phones to Wi-Fi.

Network Cable	Carries data between network devices through physical connection.	At home or in an office, connect a computer to a router using an Ethernet cable.

3. **Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.**

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Type of Network Cable	Image	Common Use	Source Link
Ethernet Cable		Used to connect computers, routers, and switches in homes, schools, and offices.	link1
Coaxial Cable		Used for cable television, broadband internet, and TV antennas.	link2
Fiber-Optic Cable		Used for high-speed internet and long-distance data transmission.	link3

4. **Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.**

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No.	Router	Switch
Main Purpose	A router connects different networks to each other.	A switch links several devices within one network.

Common Location	A router is generally installed between a local network and the internet.	A switch is commonly found in offices, schools, and computer laboratories.
Data Processing	A router directs data packets between networks using IP addresses.	A switch sends data frames to the appropriate devices using MAC addresses.
OSI Layer	A router mainly functions at Layer 3, known as the Network Layer.	A switch mainly functions at Layer 2, known as the Data Link Layer.
Data Routing	A router checks its routing table to select a suitable path for data.	A switch checks its MAC address table to send data to the correct device.
Network Communication	A router enables communication between separate networks.	A switch enables communication among devices connected to the same local network.
Example	A home router provides internet access to connected devices.	An office switch connects computers, printers, and other network devices.